

EGA Data Access Request

Application to **[DAC NAME]**

Universal Single-Cell Transcriptomic Reference
via Non-Negative Matrix Factorization

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1. Applicant Information

Principal Investigator: Zach DeBruine, PhD
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ORCID: *[ORCID ID]*
Ethics approval: *[IRB/Ethics committee reference]*

2. Datasets Requested

We request access to the following EGA studies and datasets managed by **[DAC NAME]** at **[DAC INSTITUTION]**:

[STUDY LIST — see Appendix]

For each dataset we request:

- Raw sequencing data (FASTQ files)
- Associated sample and donor metadata (cell type, tissue, disease status, age, sex)

3. Research Purpose

3.1. Scientific Justification

We are constructing a comprehensive, splicing-resolved single-cell transcriptomic reference by reprocessing raw FASTQ files from all available single-cell and single-nucleus sequencing studies into harmonized count matrices. These matrices are decomposed via **Non-Negative Matrix Factorization (NMF)** to identify universal gene expression programs that capture the full diversity of human cellular states.

The requested datasets are essential because they contain **unique tissue types, disease contexts, and patient populations** not represented in publicly accessible repositories (GEO, ENCODE). The European-origin cohorts in EGA are particularly valuable for ensuring that the reference captures population-level variation in gene programs.

3.2. Specific Use of Data

1. **Reprocessing:** Raw FASTQ files will be aligned to GRCh38 using STARsolo with splice-aware quantification, producing per-sample count matrices with spliced/unspliced layers.
2. **NMF Decomposition:** Count matrices will be integrated with our existing catalog (~180K public samples) and decomposed into a genes-by-factors matrix **W** and a factors-by-cells matrix **H**.
3. **Public Output:** Only the non-identifiable **W** matrix (gene loadings, a population-level gene-program dictionary) and k-anonymized summary statistics will be released publicly. No individual-level data is distributed.

3.3. Non-commercial Declaration

This research is conducted in a **non-commercial, academic capacity**. No individual-level data, derived products containing individual information, or sequence data will be sold, licensed, or transferred to commercial entities.

4. Data Security and Processing Environment

4.1. Processing Location

All data will be downloaded to and processed on the university's **on-premises High-Performance Computing (HPC) cluster**. No data will be uploaded to commercial cloud services.

HPC System:	<i>[Cluster name] at [University]</i>
Location:	<i>[City, Country]</i>
Data jurisdiction:	United States
Physical security:	University data center, badge access, 24/7 surveillance
Network:	Campus firewall, IDS/IPS, VPN-only access
Authentication:	University SSO + MFA; SSH key-based HPC access
Encryption:	AES-256 at rest; TLS 1.3 in transit
Access control:	POSIX ACLs; PI + approved personnel only
Audit logging:	auditd; quarterly security reviews

4.2. Data Transfer

Data will be downloaded to the university HPC via the **EGA Download Client** (pyega3) over encrypted HTTPS connections. No data will be transferred to any other system.

4.3. Data Retention and Destruction

- Raw FASTQ files will be deleted after successful alignment and QC verification.
- Count matrices will be retained only for the approved project duration.
- All data will be securely destroyed (NIST 800-88 guidelines) upon project completion, access expiration, or revocation.
- A deletion certificate will be provided to the DAC upon request.

5. Privacy and Ethical Considerations

5.1. No Re-identification

We will make no attempt to re-identify study participants from the genomic data. Metadata is used solely for annotating NMF factors (e.g., associating factors with tissue types or disease categories) at the aggregate level.

5.2. Output Privacy

The $\mathbf{W}$ matrix is a population-level parameter containing no individual data. Each column represents a gene co-expression program learned from millions of cells collectively. Removing any single donor changes $\mathbf{W}$ by $O(1/N)$, making individual contributions non-recoverable.

Summary statistics are released only after k-anonymization ($k \geq 10$ donors per group).

5.3. GDPR Considerations

As the data originates from European subjects and is processed in the United States:

- Processing is conducted under the **scientific research exemption** (GDPR Article 89).
- No individual-level data leaves the secure compute environment.
- A **Data Processing Agreement (DPA)** can be executed with the data controller if required.
- The university maintains Standard Contractual Clauses (SCCs) for EU–US data transfers as required post-Schrems II.

6. Personnel

Name	Role	Responsibilities
Zach DeBruine, PhD	PI	Study design, NMF methods, oversight
[TBD]	Co-I	Pipeline development, QC
[TBD]	Data Analyst	Reprocessing, metadata curation

All personnel have completed appropriate research ethics training and will not access data until formal approval is received from the DAC.

7. Signatures

Zach DeBruine, PhD
Principal Investigator

Date

[Signing Official Name]
[Title], [University]

Date